

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details:

Batch Number: 13

Start date: February 02, 2026

Names: Naina Anil Pandey

Internship Duration: 8 Weeks

1. Project Title

Developing an AI System for Personalized Content Creation in Media.

2. Project Objective

The main objective of this project was to design and develop an AI-powered system capable of generating personalized content based on user inputs. The system leverages Large Language Models (LLMs) to produce high-quality content such as social media posts, advertisements, professional emails, and conversational text.

This project addresses real-world challenges by:

- Automating content creation
- Reducing manual effort
- Improving productivity
- Ensuring consistency and brand relevance

Additionally, the system focuses on building a reliable and scalable solution by implementing a multi-model architecture and user-friendly interface.

3. Project description

The internship project focused on building an intelligent content generation platform (named "Manifest Core") using AI and modern full-stack web technologies. The system allows users to input prompts and select content types, after which it generates tailored and context-aware outputs.

Approach: The project was executed in multiple stages:

- Explored LLMs like Gemini and Groq
- Applied prompt engineering techniques
- Developed backend using FastAPI
- Built frontend using React, Vite, Tailwind CSS
- Implemented validation using NLP techniques

Key Features

- Personalized content generation
- Multi-model cascade system
- A/B testing
- History tracking
- Export (PDF, Word)

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project kickoff, understanding requirements, AI model research	Explored multiple LLMs (Gemini, Groq), tested APIs, understood limitations
Week 2	Prompt engineering & backend structure design	Built structured prompt templates and modular backend routing
Week 3	AI integration & backend API development	Implemented FastAPI APIs and multi-model cascade system
Week 4	Content Validation & Data Persistence implementation.	Developed NLP validation (sentiment, readability) and JSON storage
Week 5	Frontend architecture & system decoupling	Migrated to React/Vite, built landing page, enabled CORS
Week 6	UI/UX development & API integration	Designed UI with Tailwind, connected frontend with backend APIs
Week 7	Testing & debugging	Fixed bugs, optimized performance, improved UI responsiveness
Week 8	Documentation & final submission	Final report submission, PPT and project demonstration.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Internship started, project planning and requirement gathering	February 2, 2026
Prototype/First Draft	Model exploration, API testing, prompt engineering setup	Milestone 1
Backend Development	Backend development, AI integration, cascade system implemented	Milestone 2
Frontend Development	Frontend development, UI/UX design, React integration	Milestone 3
Final Submission	Final documentation & submission	Week 7 (Milestone 8)
Presentation	Final presentation and demonstration	Week 8 (Milestone 8)

5b. Project execution details

The project was completed in three phases:

- **Phase 1:** Research on AI models and prompt engineering
- **Phase 2:** Backend development using FastAPI and AI integration
- **Phase 3:** Frontend development using React and system integration

Testing and debugging were performed to ensure smooth functioning.

6. Snapshots / Screenshots

Following link contains all the relevant screenshots of the project:

https://drive.google.com/drive/folders/189xUEMbFbILdDKooeot4W_FB6qFpcMY8

7. Challenges Faced

- **API Rate Limiting:** Free-tier LLMs frequently timed out. This was resolved by engineering a High-Availability Fallback Mechanism in Python to automatically reroute traffic to secondary models.
- **Client-Server Communication:** Transitioning to a decoupled architecture caused CORS (Cross-Origin Resource Sharing) blocks. This was mitigated by configuring flask-cors middleware.
- **State Management:** Handling complex user inputs in React required careful implementation of useState and useEffect hooks, alongside local storage for session persistence.
- **Authentication handling:** Implemented Firebase + JWT

8. Learnings & Skills Acquired

Technical Skills

- Python & FastAPI
- React & Tailwind CSS
- API integration
- Prompt engineering

Soft Skills

- Problem-solving
- Debugging
- Time management

9. Testimonials from team

Building the Manifest Core engine was a profound journey into full-stack AI development. Abstracting the complexity of multiple LLMs into a single, intuitive interface taught me the critical balance between powerful backend engineering and accessible user design. Witnessing the cascade engine successfully fall back during API limits without breaking the UI—and transforming a basic script into a highly animated, React-based SaaS platform—was an incredible technical victory that prepared me for enterprise-level development.

10. Conclusion

The "Developing an AI System for Personalized Content Creation in Media" project successfully culminated in a production-ready SaaS prototype. By migrating from a basic Streamlit script to a fully decoupled React and Flask architecture, the application bridges the gap between raw LLM capabilities and practical media generation needs. The system successfully abstracts complex prompt engineering, ensures high availability through fallback routing, and guarantees brand safety via programmatic validation—all delivered through an exceptional, highly responsive user interface. This internship provided invaluable, real-world experience in developing scalable, AI-driven web applications.

11. Acknowledgements

would like to extend my sincere gratitude to the **Infosys Springboard 6.0** program for providing this incredible platform for learning and innovation. Special thanks to my mentors and my team members for their guidance, strategic roadmaps, and support in establishing professional corporate expectations throughout this 8-week journey. Their insights were instrumental in elevating this project from a simple script to a comprehensive software platform.